

How to Add a User to a Group in Linux

 linuxize.com/post/how-to-add-user-to-group-in-linux

Dejan Panovski

August 24, 2018

In Linux, groups are used to organize user accounts and control access to files and resources. Instead of assigning permissions to each user individually, you assign them to a group and add users to it.

There are two types of groups:

- **Primary group** — assigned to a user when the account is created. Usually, the name of the primary group is the same as the username. When a user creates a file, the file's group is set to the user's primary group. The primary group is stored in the [/etc/passwd](#) file.
- **Secondary (supplementary) groups** — used to grant additional permissions. For example, adding a user to the `docker` group allows them to run Docker commands without `sudo`.

Each user has exactly one primary group and can belong to zero or more secondary groups.

Only root or users with [sudo](#) privileges can modify group membership.

Quick Reference

Task	Command
View user's groups	<code>id username</code>
List group names only	<code>groups username</code>
List all system groups	<code>getent group</code>
List members of a group	<code>getent group groupname</code>
Add user to a group	<code>sudo usermod -a -G groupname username</code>
Add user to multiple groups	<code>sudo usermod -a -G group1,group2 username</code>
Change primary group	<code>sudo usermod -g groupname username</code>
Create user with groups	<code>sudo useradd -g primary -G sec1,sec2 username</code>
Remove user from a group	<code>sudo gpasswd -d username groupname</code>
Create a group	<code>sudo groupadd groupname</code>
Delete a group	<code>sudo groupdel groupname</code>

Displaying User Groups

Before modifying groups, it is useful to check a user's current group membership.

The `id` command displays the user's UID, primary group, and all secondary groups:

```
uid=1000(linuxize) gid=100(users)  
groups=100(users),10(wheel),95(storage),98(power),990(libvirt),993(docker),999(kvm)
```

The `groups` command prints only the group names:

```
wheel storage power users libvirt docker kvm
```

If you omit the username, both commands display information for the currently logged-in user.

Listing All Groups on the System

To list all groups on the system:

To list all members of a specific group:

```
docker:x:993:linuxize,deploy
```

Adding a User to a Group

To add an existing user to a secondary group, use the `usermod -a -G` command followed by the group name and the username:

```
sudo usermod -a -G groupname username
```

For example, to add the user `linuxize` to the `sudo` group:

```
sudo usermod -a -G sudo linuxize
```

Always use the `-a` (append) option when adding a user to a group. If you omit `-a`, the user will be **removed from all secondary groups** not listed after the `-G` option.

On success, the command produces no output. It warns you only if the user or group does not exist.

Group changes apply after the user logs out and logs back in. To apply the change in the current session, run:

Adding a User to Multiple Groups

To add a user to multiple secondary groups at once, separate the group names with commas (no spaces):

```
sudo usermod -a -G group1,group2,group3 username
```

Changing a User's Primary Group

To change a user's primary group, use `usermod` with the lowercase `-g` option:

```
sudo usermod -g groupname username
```

For example, to change the primary group of the user `linuxize` to `developers`:

```
sudo usermod -g developers linuxize
```

Note the difference: `-G` (uppercase) sets secondary groups, `-g` (lowercase) sets the primary group.

Creating a User with Groups

The [useradd](#) command can assign both a primary and secondary groups when creating a new user:

```
sudo useradd -g users -G wheel,developers nathan
```

This creates a user `nathan` with `users` as the primary group and `wheel` and `developers` as secondary groups.

Removing a User from a Group

To remove a user from a group, use the `gpasswd` command with the `-d` option:

```
sudo gpasswd -d username groupname
```

Creating and Deleting Groups

To [create a new group](#) :

To [delete a group](#) :

You cannot delete a group that is a user's primary group. Change the user's primary group first.

FAQ

What happens if I forget the **-a** flag with **usermod -G**?

The user will be removed from all secondary groups except those listed in the command. This is the most common mistake when managing groups. Always use **-a -G** together.

Do group changes take effect immediately?

No. The user must log out and log back in for the new group membership to take effect. You can verify by running **id username** — the change shows up there immediately, but the user's active session still uses the old groups.

What is the difference between **-g** and **-G** in **usermod**?

Lowercase **-g** sets the user's primary group. Uppercase **-G** sets secondary (supplementary) groups. When used with **-a**, **-G** appends to the existing list instead of replacing it.

How do I find out which groups exist on the system?

Run **getent group** to list all groups, or **getent group groupname** to check whether a specific group exists and see its members.

Can a user belong to multiple primary groups?No. Each user has exactly one primary group. To grant access to multiple resources, use secondary groups.

Conclusion

The **usermod -a -G** command adds users to secondary groups in Linux. Use **id** or **groups** to check membership, **gpasswd -d** to remove a user from a group, and always include the **-a** flag to avoid removing existing group assignments.

The same commands apply to any Linux distribution, including Ubuntu, CentOS, RHEL, Debian, and Linux Mint.

If you have any questions, feel free to leave a comment below.